Calculus II

Tutorial #9 14–18 Apr 2025

Topics:

- A simple population model
- First-order linear differential equations
- Change of variables (Homogeneous first-order linear differential equations; Bernoulli's equations)

• **True-False Review**. Decide if the given statement is **True** or **False**, and justify your answer.

1. (a) There is a unique integrating factor for a differential equation of the form

$$y' + p(x)y = q(x).$$

(b) An integrating factor for the differential equation

$$\frac{dy}{dx} = x^2 y + \sin x$$

is

$$I(x) = e^{\int x^2 \, dx.}$$

2. The change of variables $y = xV(x)$ always turns a first-order homogeneous differential equation into a separable differential equation for V as a function of x.

3. The following are Bernoulli differential equations

(a)

$$\frac{dy}{dx} = \sqrt{x}y + \sqrt{xy}$$

(b)

$$\frac{dy}{dx} - e^{xy}y = 5x\sqrt{y}$$

• **Problems**.

4. A hot metal bar whose temperature is 350°F is placed in a room whose temperature is constant at 70°F. After 2 minutes, the temperature of the bar is 210°F. Using Newton's law of cooling, determine

(a) the temperature of the bar after 4 minutes

(b) the time requires for the bar to cool to 100°F.

5. The number of bacteria in a culture grows at a rate proportional to the number present. After 10 hours, there were $5,000$ bacteria present, and after 12 hours, $6,000$ bacteria. Determine the initial size of the culture and the doubling time of the population.

6. Solve the following differential equations.

   (a) $\dfrac{dy}{dx} + xy = xe^{x^2/2}$

   (b) $(t-1)\dfrac{dx}{dt} - x - t = 0$

7. Solve the given IVP

$$\frac{dy}{dx} + \frac{2x}{1+x^2}y = xy^2, \quad y(0) = 1.$$

8. Determine whether the given function is homogeneous of degree zero. Rewrite those that are as functions of the single variable $V = \dfrac{y}{x}$.

   (a) $f(x,y) = \dfrac{x^2 - y^2}{xy}$

   (b) $f(x,y) = \dfrac{y}{x-1}$

9. Solve the following differential equations.

   (a) $(3x - 2y)\dfrac{dy}{dx} = 3y$

   (b) $\sin\left(\dfrac{y}{x}\right)(xy' - y) = x\cos\left(\dfrac{y}{x}\right)$

10. Identify the following differential equations are Bernoulli equations and solve them.

    (a) $y' - x^{-1}y = 4x^2y^{-1}\cos x,\ x > 0.$

    (b) $\dfrac{dy}{dx} - \dfrac{3}{2x}y = 6y^{\frac{1}{3}}x^2\ln x.$

END

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Answers to selected problems

1. True 2. False 3. True

4. $140°F, \quad \approx 6.4$ minutes

5. ≈ 7.6 hours

6. (a) $y = e^{-x^2/2}\left(\frac{1}{2}e^{x^2} + C\right)$ (b) $x = (t-1)\left(\ln|t-1| - \frac{1}{t-1} + C\right)$

7. $y = \frac{1}{(1+x^2)\left(-\frac{1}{2}\log(1+x^2)+1\right)}$

8. (a) Yes (b) No

9. (a) $y = 0$ and $y = Ce^{-\frac{3x}{2y}}$ (b) $x\cos\left(\frac{y}{x}\right) = C$

10. (a) $y^2 = x^2(8\sin x + C)$ (b) $y = 0$ and $y^{\frac{2}{3}} = x[x^2(2\ln x - 1) + C]$.